

ThreatWatch Lite

Endpoint Threat Detection Dashboard

Documentation generated on: 2025-05-01

ThreatWatch Lite is a minimalist security dashboard designed for monitoring endpoint threats. It uses pattern matching and NLP-based analysis to detect potential security issues in log files.

Table of Contents

- 1. Introduction
- 2. Dashboard Overview
- 3. Log Upload Functionality
- 4. Threat Analysis & Detection
- 5. Detailed Threat View
- 6. Filtering & Categorization
- 7. API Integration
- 8. Technical Architecture

1. Introduction

ThreatWatch Lite is a cybersecurity dashboard focused on endpoint threat detection. It provides a streamlined, minimalist interface for monitoring and analyzing security threats detected in system logs.

Key Features:

- NLP-based threat detection using pattern matching and keyword analysis
- Real-time threat scoring and categorization
- Support for manual log uploads and simulated agent feeds
- Detailed threat analysis with remediation suggestions
- API endpoints for integration with other security tools

This application focuses on a lightweight, pattern-matching approach to threat detection rather than using external AI services, making it suitable for environments with limited connectivity or privacy requirements.

2. Dashboard Overview

The main dashboard provides an at-a-glance view of all detected threats, organized by severity level (High, Medium, Low). The interface uses a minimalist design with contextual colors to highlight critical information.

[Screenshot: Fig 1: ThreatWatch Lite Dashboard]

Dashboard Components:

- Threat Table: Displays all detected threats with severity, categories, and detected keywords
- Upload Panel: Allows users to submit log files for analysis or paste raw log content
- Quick Filters: Enables filtering by threat level, keywords, or categories
- API Endpoints: Documentation for integrating with the system programmatically

Color Coding:

- Red: High severity threats that require immediate attention
- Yellow: Medium severity threats that should be investigated
- Green: Low severity threats that are less urgent

3. Log Upload Functionality

ThreatWatch Lite supports two methods for submitting log data: direct file upload or pasting raw log content. Both methods trigger the same analysis pipeline.

[Screenshot: Fig 2: Log Upload Interface]

Upload Methods:

- File Upload: Accepts .log and .txt files containing system or application logs
- Text Input: Allows pasting raw log content directly into the text area
- API Submission: Logs can be submitted programmatically via the /upload endpoint

Supported Sources:

- Manual uploads from security analysts
- Simulated agent feeds (for testing/demonstration)
- Potential integration with real endpoint agents

4. Threat Analysis & Detection

ThreatWatch Lite uses a combination of pattern matching, keyword detection, and contextual analysis to identify potential security threats in log data.

Detection Process:

1. Log Scanning: Each log is scanned for known threat patterns and suspicious keywords
2. Pattern Matching: Detected patterns are matched against a database of known threats
3. Contextual Analysis: The system analyzes surrounding context to improve detection accuracy
4. Threat Scoring: A numerical score is calculated based on the severity and number of detected patterns
5. Categorization: Threats are categorized into types (e.g., `command_execution`, `credential_theft`)

NLP Techniques Used:

- Keyword frequency analysis
- Pattern recognition for command structures
- Contextual indicators (unusual time patterns, repeated failures)
- Correlation between multiple suspicious activities

5. Detailed Threat View

When a threat is detected, users can access a detailed view that provides comprehensive information about the threat, including raw logs, detected patterns, and potential security implications.

[Screenshot: Fig 3: Detailed Threat Analysis View]

Detailed View Components:

- *Threat Summary: ID, timestamp, source, and severity level*
- *Detected Keywords: Specific suspicious patterns found in the log*
- *Threat Categories: Types of security threats identified*
- *Raw Log Display: Complete log entry with highlighted suspicious elements*
- *Security Advisory: Potential impact and recommended actions*
- *MITRE ATT&CK Mapping: Correlation with known attack techniques*

6. Filtering & Categorization

ThreatWatch Lite provides robust filtering capabilities to help security analysts focus on the most relevant threats and patterns.

[Screenshot: Fig 4: Threat Filtering Interface]

Filtering Capabilities:

- *By Severity Level: Filter to show only High, Medium, or Low threats*
- *By Keyword: Search for specific suspicious commands or patterns*
- *By Category: Filter by threat categories (e.g., 'command_execution', 'credential_theft')*
- *Time-based: Focus on recent threats (most recent shown at the top)*

Categorization System:

ThreatWatch Lite automatically categorizes threats into groups that help analysts understand the nature of potential attacks:

- *command_execution: Suspicious command or script execution*
- *credential_theft: Attempts to access or steal authentication credentials*
- *remote_access: Unusual remote access attempts or patterns*
- *privilege_escalation: Attempts to gain higher privileges*
- *data_exfiltration: Potential data theft or unauthorized transfers*
- *persistence: Actions that maintain presence on compromised systems*

7. API Integration

ThreatWatch Lite provides a simple RESTful API that enables integration with other security tools and automation workflows.

Available Endpoints:

1. POST /upload

Purpose: Submit log data for analysis

Payload: {"log": "log data here", "source": "endpoint_name"}

Response: JSON object with threat details if detected

2. GET /api/threats

Purpose: Retrieve all detected threats

Parameters: None

Response: JSON array of all threat objects

3. GET /api/threat/:id

Purpose: Get detailed information about a specific threat

Parameters: Threat ID in URL

Response: JSON object with complete threat details

4. GET /filter_threats

Purpose: Filter threats by level or keyword

Parameters: ?level=High&keyword=powershell&category=command_execution

Response: JSON array of matching threat objects

8. Technical Architecture

ThreatWatch Lite is built with a modular architecture that separates concerns and allows for easy extension and maintenance.

Component Architecture:

1. Flask Web Application (main.py, app.py, routes.py)

- Handles HTTP requests and routes
- Manages sessions and user interface
- Processes form submissions and API calls

2. Database Layer (models.py)

- SQLAlchemy ORM for database interactions
- Threat model for storing detected threats
- JSON serialization for API responses

3. Threat Detection Engine (threat_detector.py)

- Pattern matching and keyword detection
- Threat scoring algorithm
- Category mapping logic

4. NLP Analysis Module (nlp_analyzer.py)

- Text analysis for threat context
- Indicator extraction from log content
- Severity calculation based on patterns

5. Frontend (templates/, static/)

- Responsive Bootstrap-based interface
- Client-side filtering and sorting

- Interactive threat visualization